



**WHY DO WE NEED
A CLEANROOM?**

DEPARTMENT OF

SPACE

P H Y S I C S



EXCELLENT SCIENTIFIC LAB at EUROPEAN LEVEL



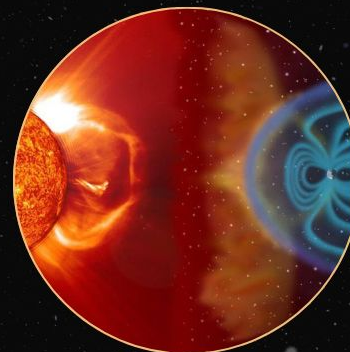
SPACE MISSIONS

Delivery of 25+ scientific instruments for space missions



UNIQUE DATA

More than 50 years at Lomnický štít cosmic rays observatory



AI driven SCIENCE

Exploring space phenomena by state-of-the-art technologies



What do we learn today?

1. Cleanroom - what is it, how does it work, and why is it necessary in the first place?
2. Our heritage - mission history and the Cleanroom Predecessor
3. How did our first Space Cleanroom come to be?

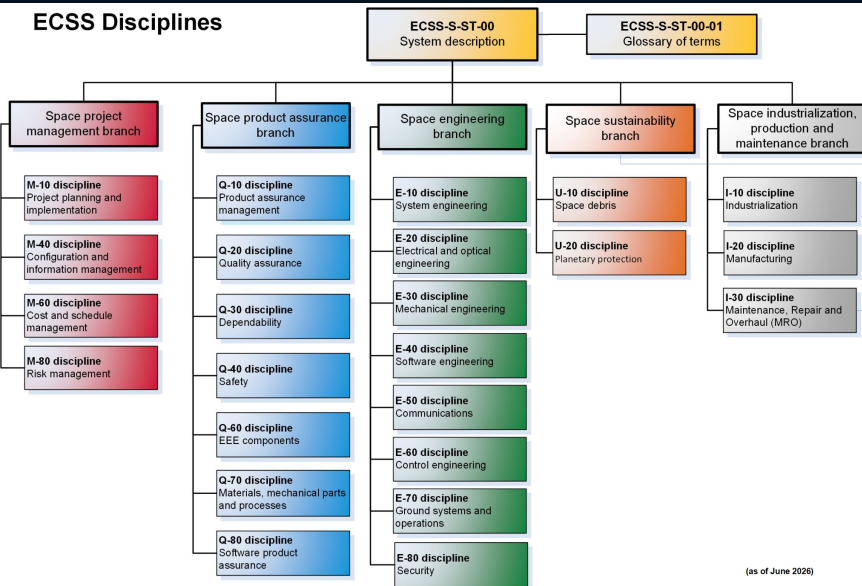


Not just any regulations

- Unification
- Integrity
- Reliability
- ecss.nl

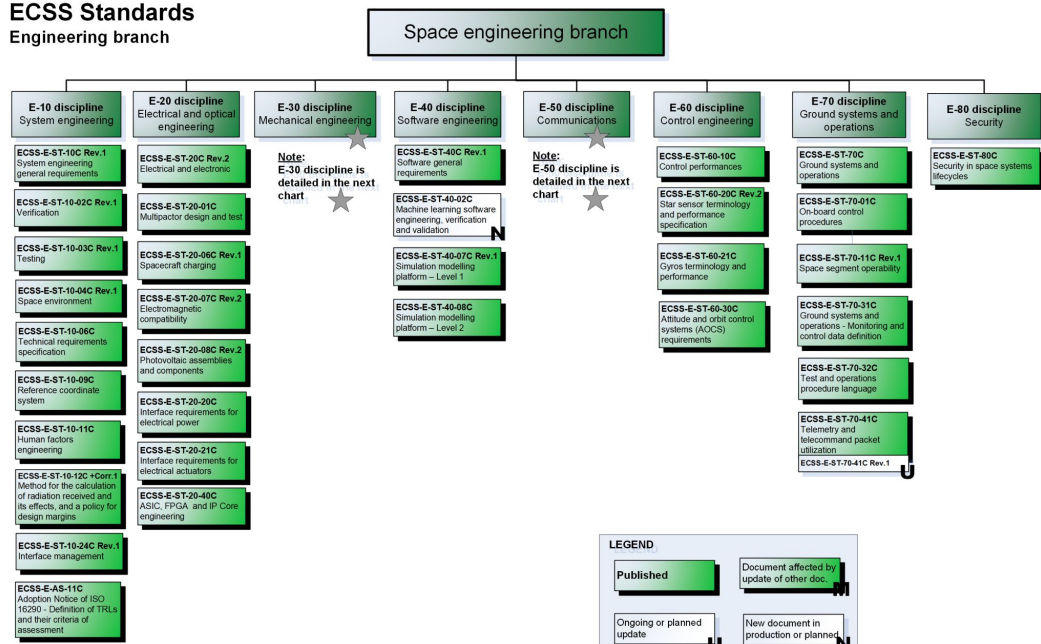
ECSS standards

ECSS Disciplines



(as of June 2026)

ECSS Standards Engineering branch



LEGEND

Published

Document affected by update of other doc.

Ongoing or planned update

New document in production or planned

(as of October 2025)

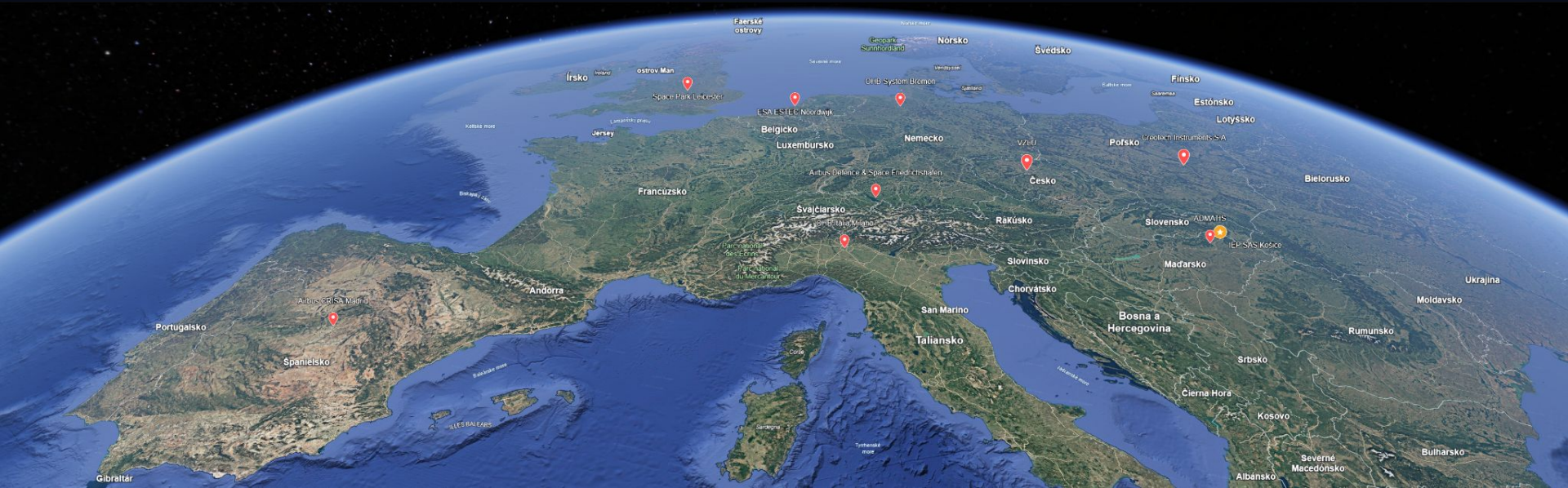


**A cleanroom is not a room that looks clean. It is
a system that keeps contamination controlled,
measured and traceable.**



"Nearby" Space Cleanroom overview

- ESA, Airbus, OHB, VZLU – top notch AIT centres
- ISO 8/7 + local ISO 5 environments
- Distance = logistical and contamination risk
- Limited access (planning months ahead)



Contamination is more than dust

WHAT CAN ENTER

- Particles and fibres
- Molecular films and residues
- Foreign material when mission-relevant

WHERE IT COMES FROM

- People and garments
- Tools, packaging and materials
- Surfaces, airflow and operations



A tiny contaminant can become a mission-level failure



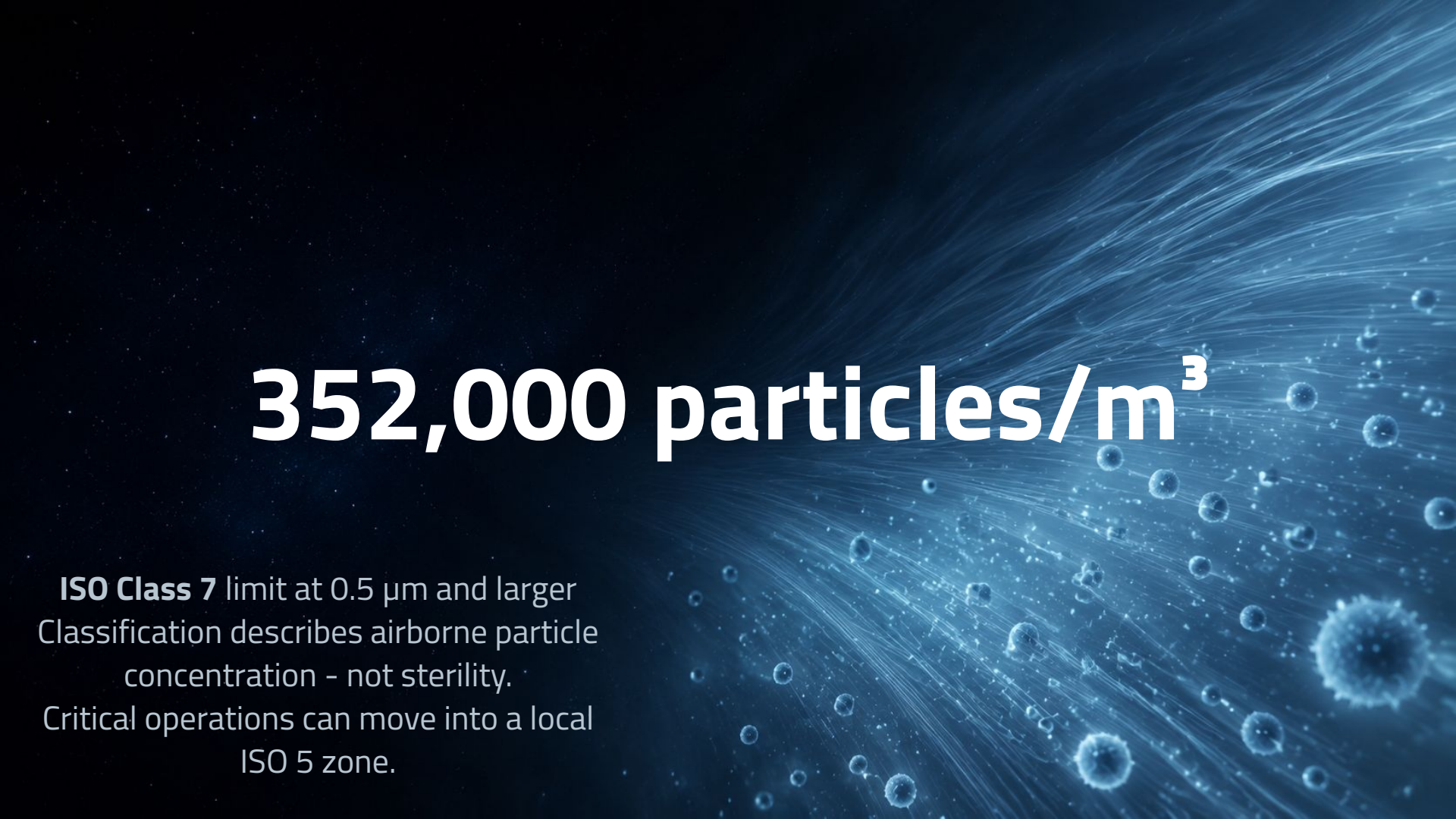
- **Optics:** lower transmission or higher stray light
- **Thermal surfaces:** changed absorptivity and emissivity
- **Mechanisms:** friction, blockage or wear
- **Electronics and detectors:** leakage, connector faults or corrupted measurement

The risk is small in size - not so small in consequence.

Characteristics

- 25 m² ISO Class 7 area
- Local ISO Class 5 zone (laminar box)
- Controlled environmental conditions (temperature / humidity / particle concentration)
- Focus on sensitive and critical electronics for space applications (development, testing and integration under ESD-controlled conditions)
- Personnel access through an air shower
- Material pass-through: 60 x 60 cm
- 30 - 60 air exchanges per hour
- Airflow rate >2000 m³/h
- Retained to introduced air ratio - 9:1



The background of the slide is a dark blue gradient with a complex, abstract pattern of glowing blue lines and numerous small, translucent blue spheres of varying sizes, resembling particles or microorganisms in motion.

352,000 particles/m³

ISO Class 7 limit at 0.5 μm and larger
Classification describes airborne particle
concentration - not sterility.
Critical operations can move into a local
ISO 5 zone.



Equipment

- Professional soldering workstations and inspection systems:
- 3D stereo digital microscope DRV-Z1
- Ersascope X Inspection System
- Complete laboratory equipment (oscilloscopes, power supplies, precision multimeters, function generator, counter, ...)
- Precision analytical balances
- Vibration testing system (Brüel & Kjær LDS V780):
- Frequency range up to 4,000 Hz
- Excitation force (sine peak) up to 5,100 N
- Acceleration (sine peak) up to $1,050 \text{ m}\cdot\text{s}^{-2}$
- Chronos 2.1-HD high-speed camera
- Thermal-vacuum chamber (Vakuum Praha SPACEVAC):
- Temperature range: $-100 \text{ }^{\circ}\text{C}$ to $+160 \text{ }^{\circ}\text{C}$
- Vacuum: $1 \times 10^{-7} \text{ mbar}$
- NI measurement modules (temperature, voltage, digital inputs/outputs and relays) for test automation
- Extensive DUT connectivity (SpaceWire, MIL-STD-1553, Ethernet, CAN and RS-485/422)

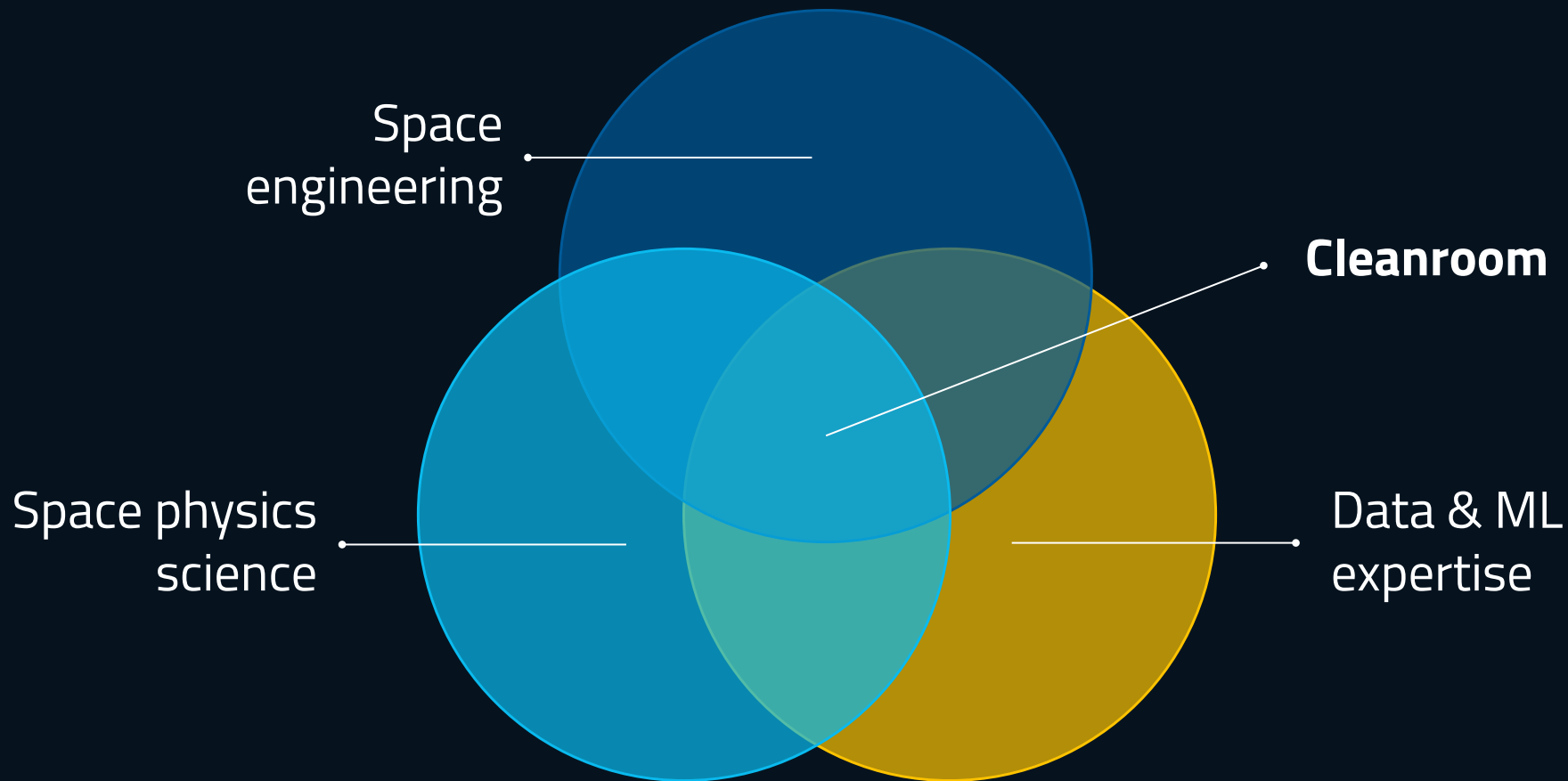
Control works only as a system

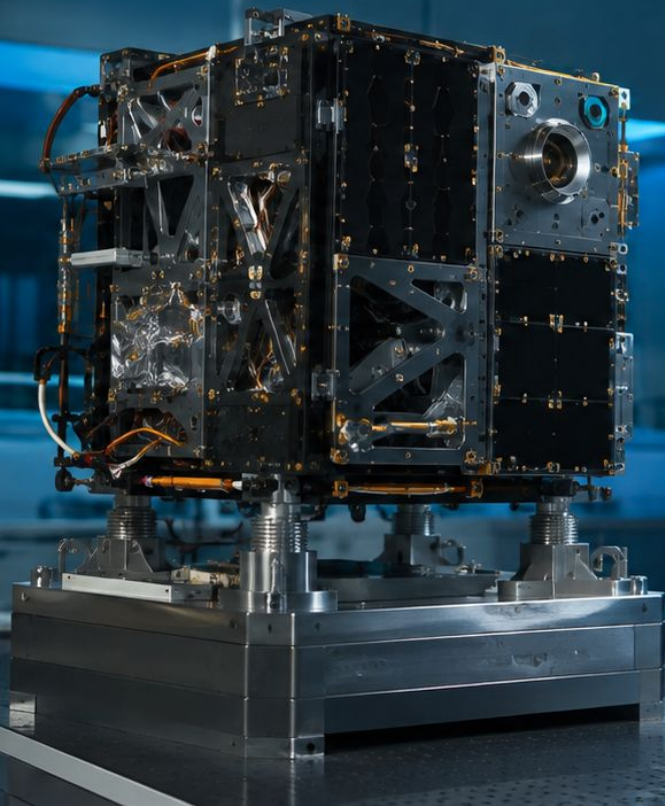
- Filtered airflow and controlled air exchange
- Pressure cascade and interlocked access
- Temperature, humidity and ESD control
- Gowning, approved materials and cleaning
- Particle monitoring, records and verification

People are usually the largest contamination source

WHAT CHANGES THE RESULT

- **Dress correctly:** gown, hood, gloves and footwear
- **Move deliberately:** speed disturbs airflow and releases particles
- **Control materials:** clean tools, approved packaging, no cardboard
- **Protect the process:** repeatable steps and records





**Clean assembly is
the first link
in a chain of
evidence**

SPACE::LAB connects preparation, inspection and testing

CONTROLLED WORK

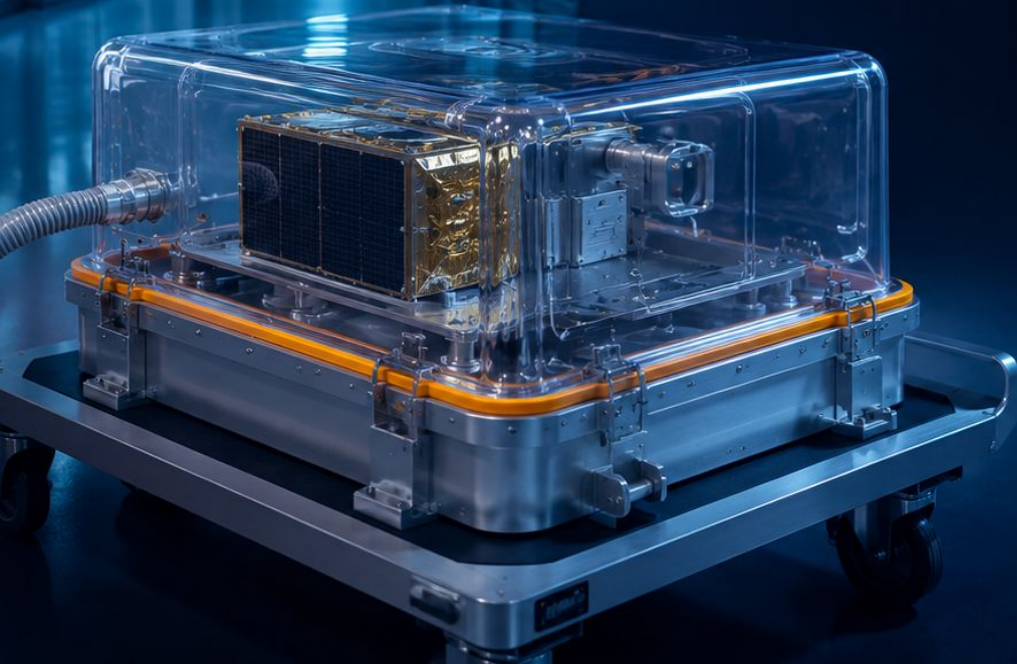
- ISO 7 integration zone
- Local ISO 5 protection
- ESD-aware workstations
- Microscopy and inspection

VERIFICATION SUPPORT

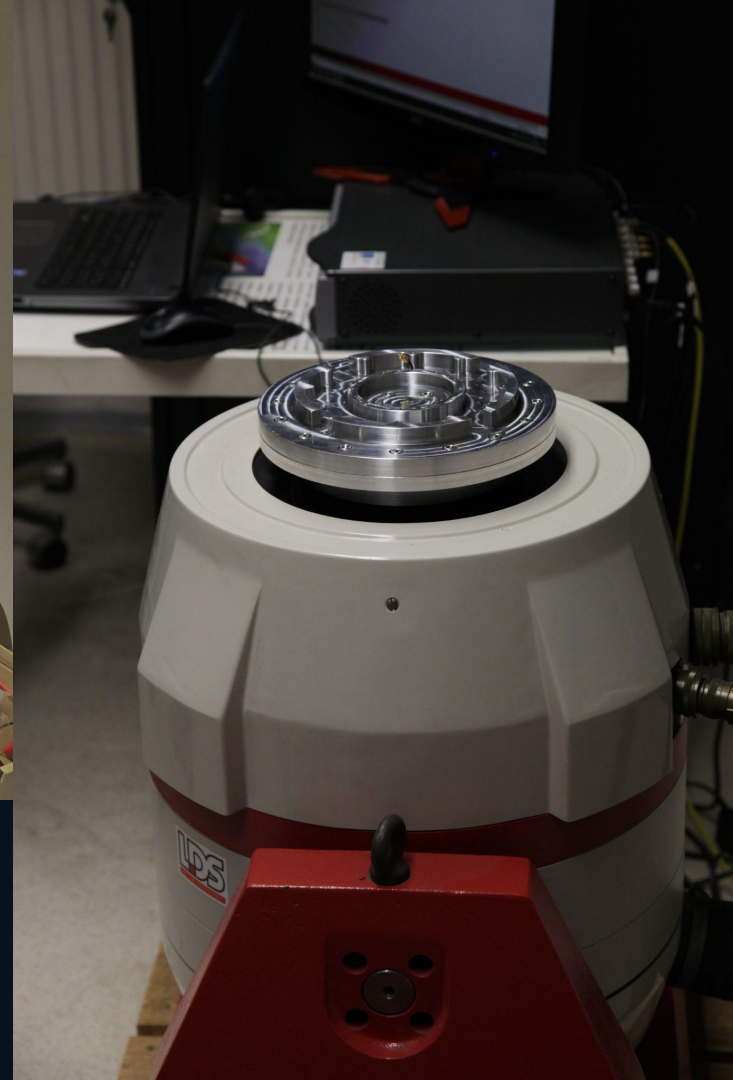
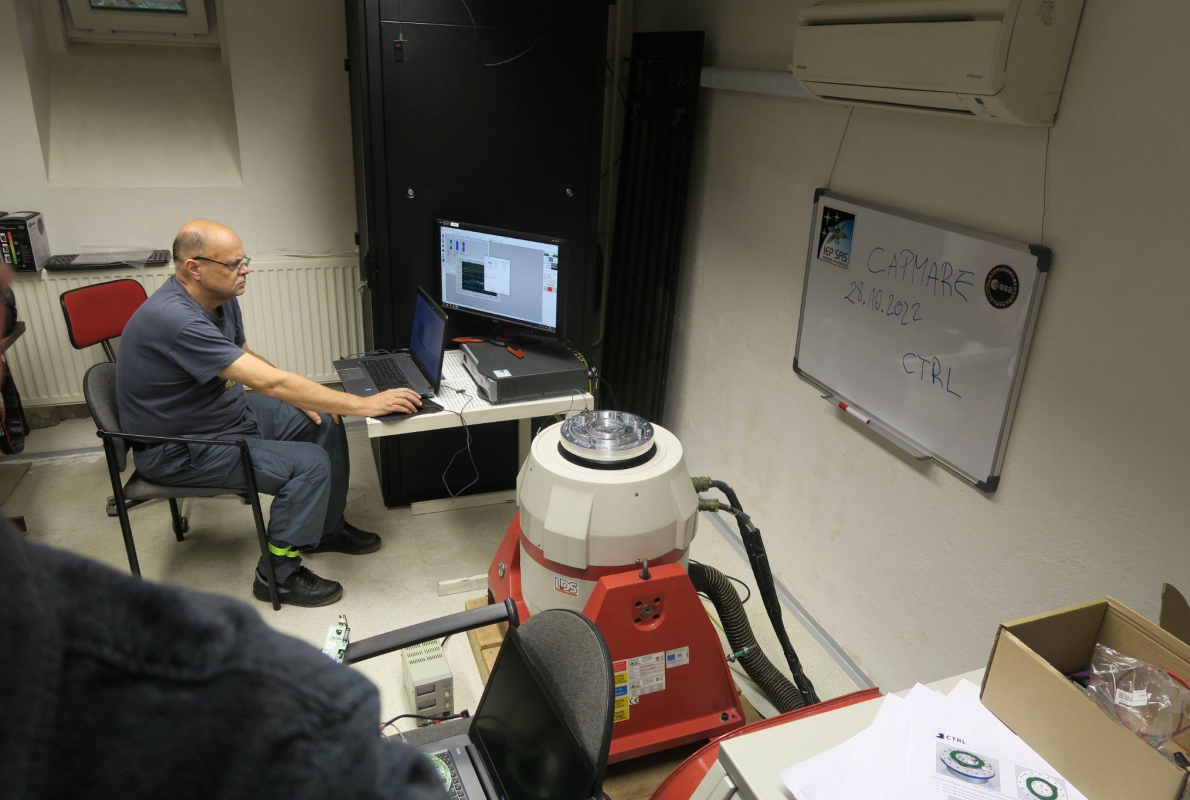
- Particle monitoring and traceability
- Mechanical vibration testing
- Thermal-vacuum testing in SPACEVAC
- DAQ, automation and measurement support



Mechanical vibration testing



- Recreates launch-induced sine and random dynamic loads
- Reveals resonances, weak joints and structural margins
- Requires a verified fixture, calibrated sensors and defined limits
- Test success means both survival and acceptable post-test performance



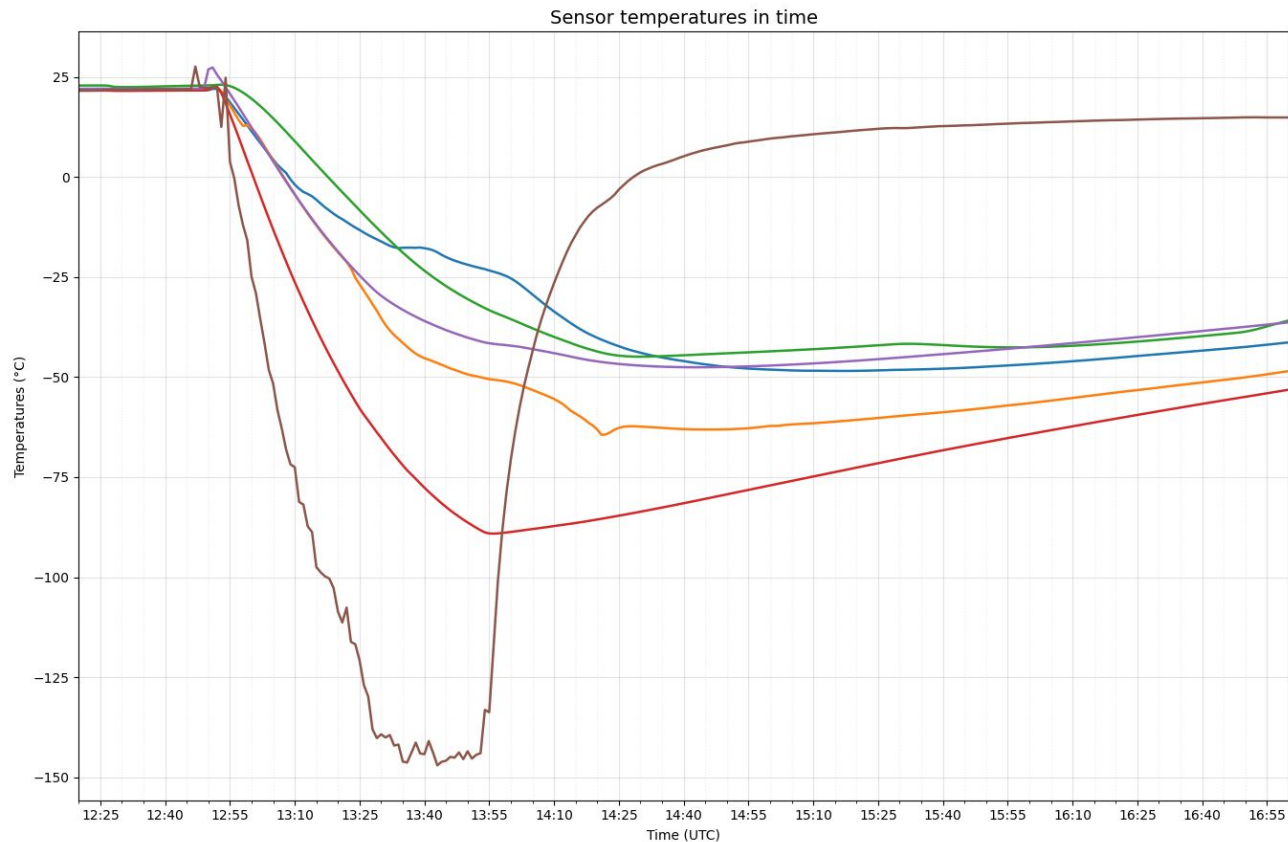
2022 - Project CAPMARE - mechanical stress tests | Ing. Ján Baláž, PhD.

Thermal-vacuum testing

- Combines low pressure with controlled hot and cold conditions
- Exposes thermal design, interface and outgassing problems
- Requires vacuum-compatible materials and a clean chamber baseline
- Uses controlled pump-down, monitoring, venting and documented results



Thermal-vacuum testing



- 1 - Copper plate base, next to EMONs
- 2 - Copper plate base, next to EMONs (position 2)
- 3 - Top of the EMONs
- 4 - Copper plate base, next to EMONs (position 3)
- 5 - Sensor beneath the copper plate
- RAD - LN2 exhaust fume (coming into the chamber)



Cleanliness and qualification as one engineering chain

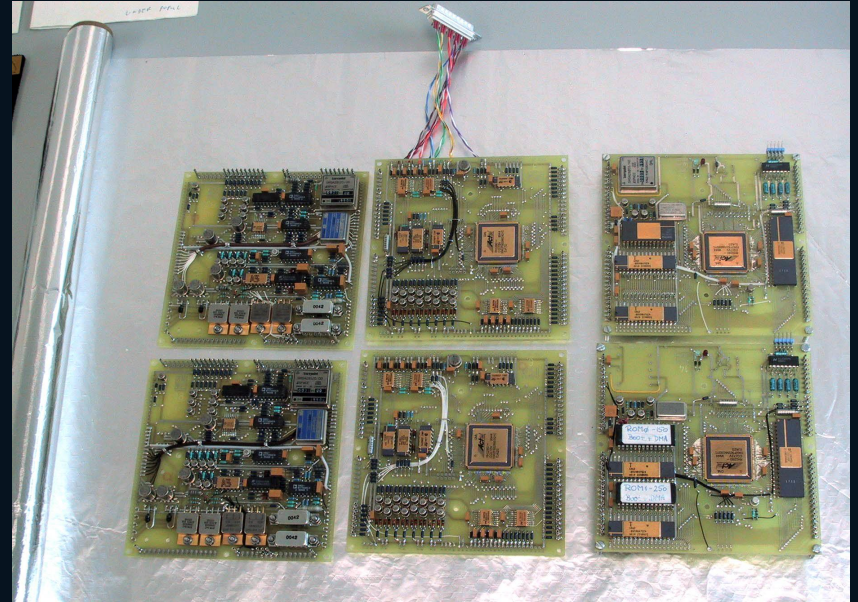
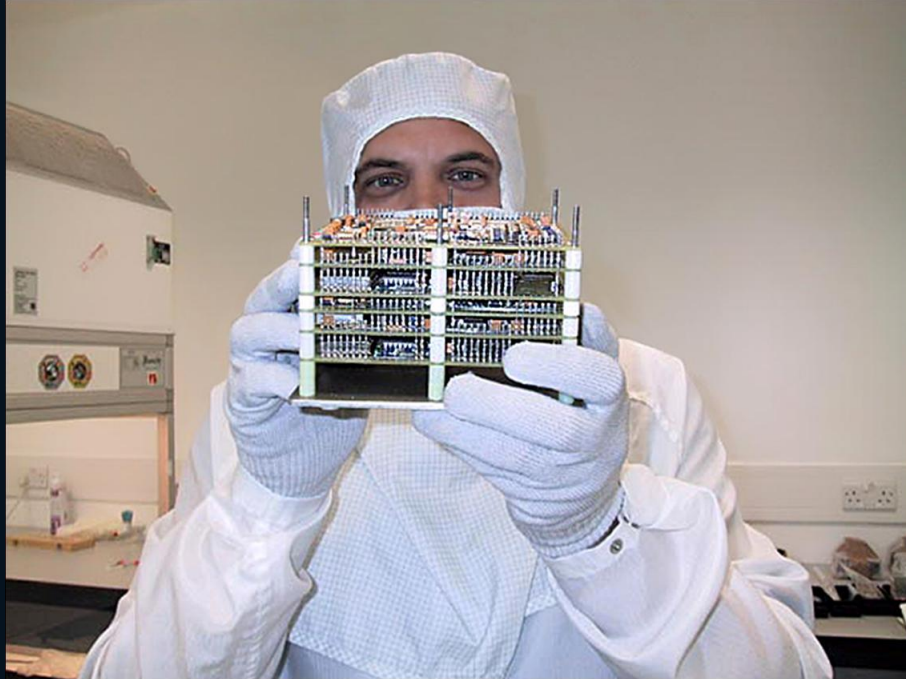
- Define sensitivity and contamination requirements
- Plan clean integration, handling and material compatibility
- Assemble and inspect under controlled conditions
- Run environmental tests without recontaminating the hardware
- Verify configuration, cleanliness and performance before release



**The facility is new.
The experience behind it is
not.**

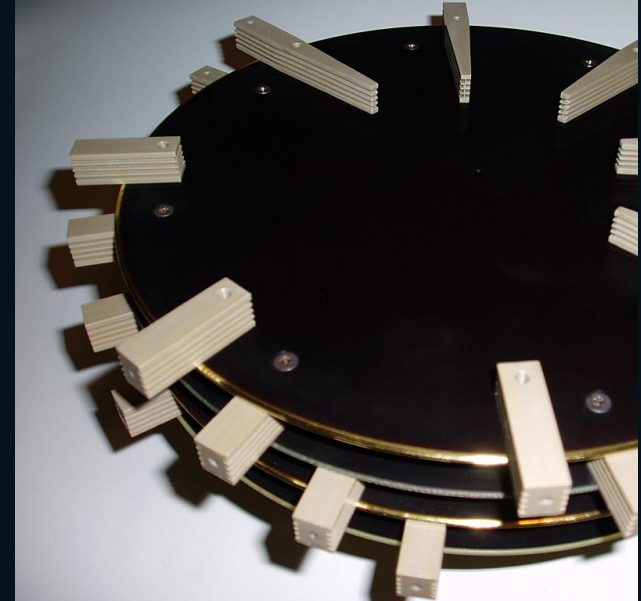
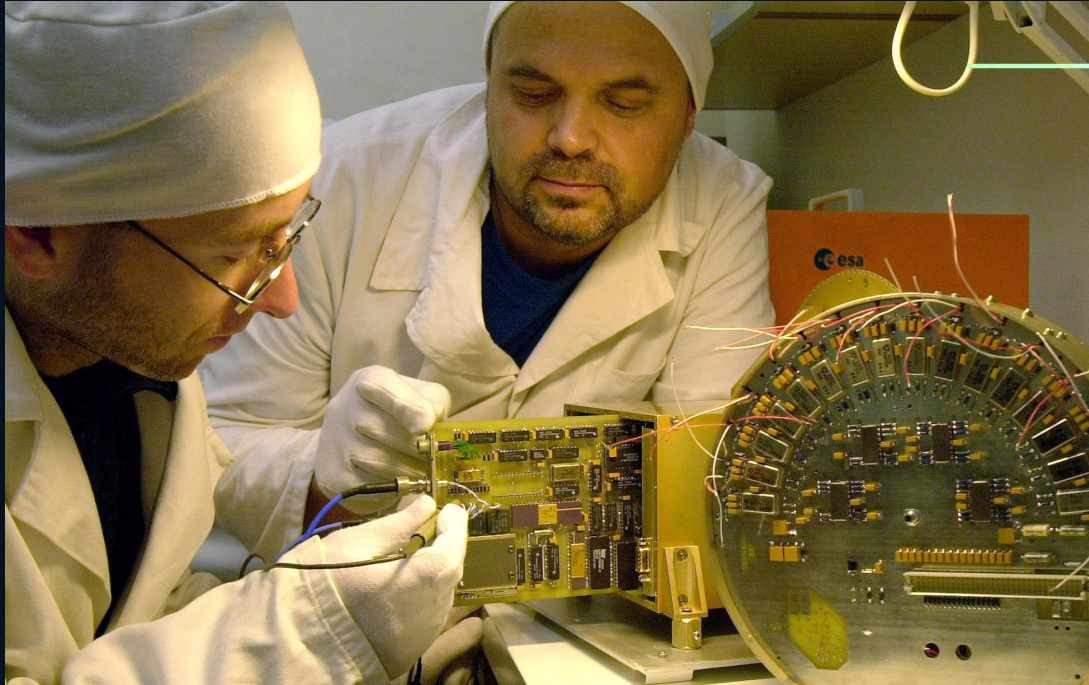


Cleanroom capability built on mission heritage...



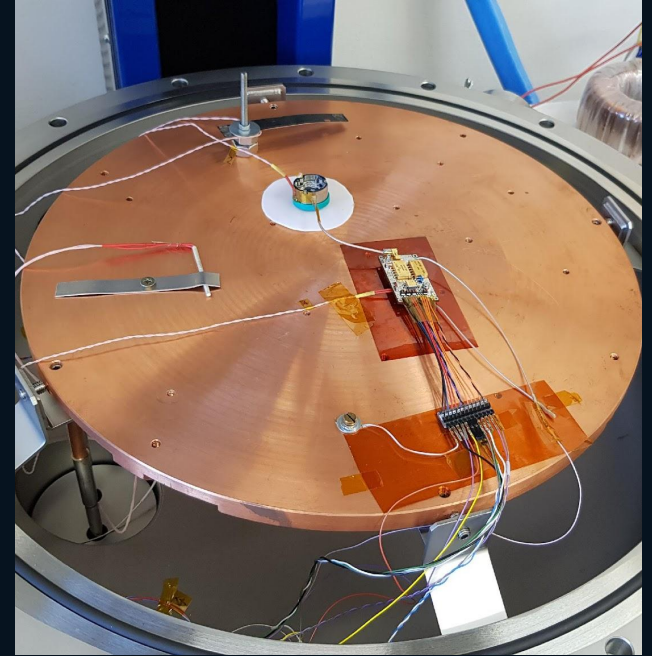
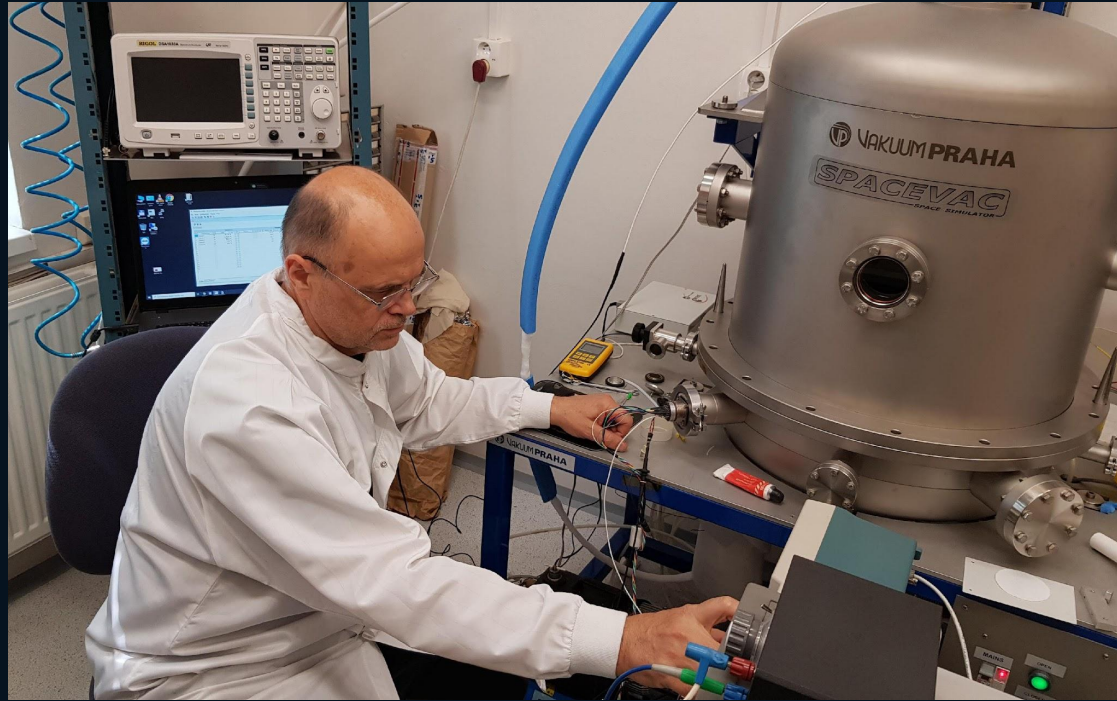
2001 - Cleanroom work in Ireland - early mission hardware
experience: ESS - ESA Rosetta | Ing. Ján Baláž, PhD.

Cleanroom capability built on mission heritage...



2003 - Cleanroom pathfinder in Košice - controlled integration and collaboration:
NUADU / ESA Double Star | Ing. Ján Baláž, PhD. & Ing. Igor Strhářský

Cleanroom capability built on mission heritage...



2020 - Thermal-vacuum tests in Košice - a direct bridge to local test capability: ESA JUICE | Ing. Ján Baláž, PhD.

**From an ordinary
room to a
controlled
space-hardware
facility**



Towards cleanroom conditions ...

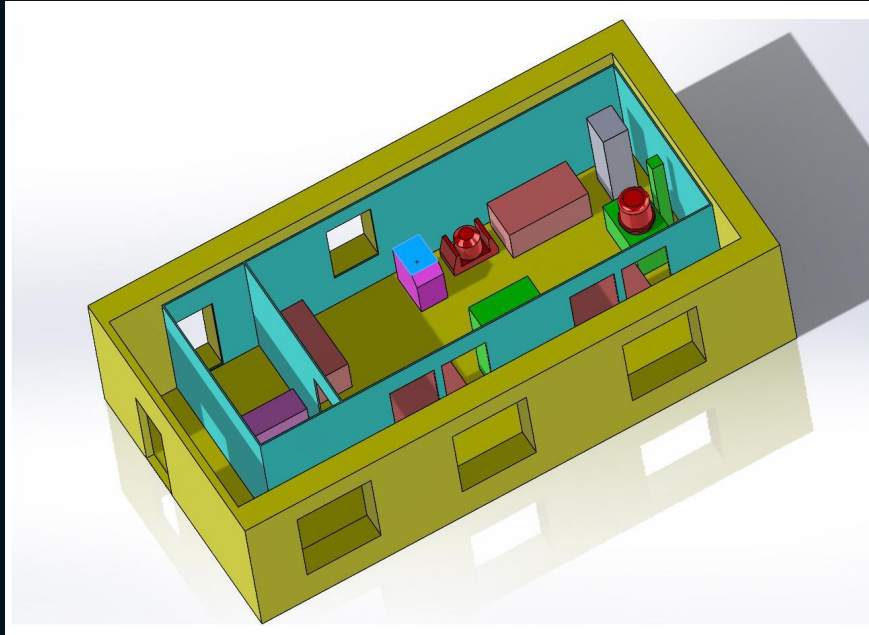


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Towards cleanroom conditions ...



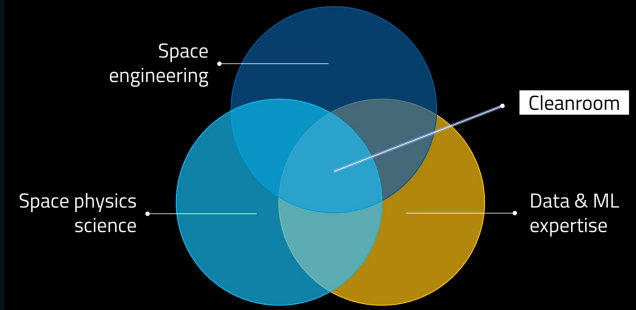
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2026-05

Cleanroom utilization

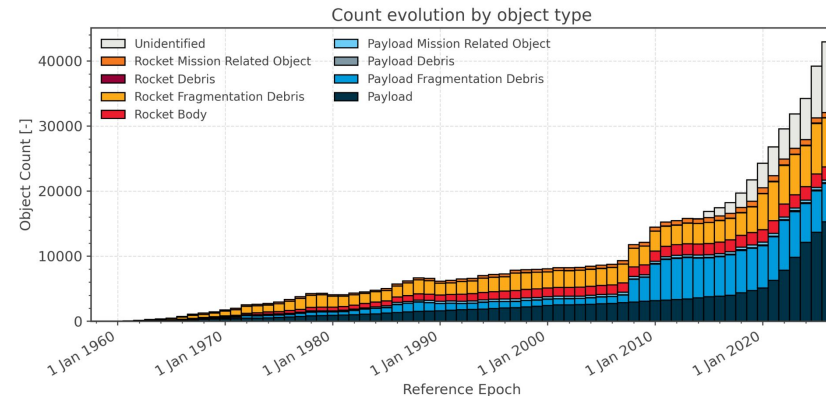
- A. Our space program
- B. Cooperation on joint scientific / technological projects
- C. Commercial rent



STEPHANIK

Slovak Technical Ecosphere Platform

Objects in Orbit





What to observe during the visit

- Airflow direction & pressure zoning
- Gowning room, air shower and material pass-through
- ISO 5 laminar-flow working area
- ESD workstations, inspection and particle monitoring
- Stations of integration, vibration and vacuum-test activities



**THANK YOU FOR YOUR ATTENTION
NEXT: CLEANROOM VISIT**